Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/21/2016
PI Name:	Hirsa, Amir Ph.D.		
Project Title:	Amyloid Fibril Formation in Microgravity: Distinguishing Interfacial and Flow Effects		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lopez, Juan (Arizona State University)		
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	Formation of amyloid fibrils is widely studied because of their role in neurodegenerative diseases, including Alzheimer's. The fibrillization process is characterized by a nucleation step followed by fibril growth, much like the steps that occur during protein crystallization. Aside from the general interest in macromolecular biophysics, recent evidence shows the relevance of amyloid fibril formation in animal models exposed to cosmic radiation, with obvious implications for interplanetary travel. A preponderance of evidence shows that agitating protein solutions can significantly accelerate amyloid fibril formation; however, the mechanisms by which the presence of bubbles and flow independently affect fibrillization are not known. The situation is further complicated because we do not yet have mechanistic models that can quantify the influence of natural convection due to gravity on the fibrillization process. We know that fluid interfaces (e.g., air-water interface) induce amyloid fibrils when they are sheared, but the influence of container walls is not fully understood.		

Task Description:	Here, we aim to distinguish the influence of flow and fluid interfaces on amyloid formation by removing other effects, including solid walls and buoyancy effects. Formation of protein amyloid fibrils provides a model system for studying biomolecular transport phenomena, where the influence of flow can be altogether negated by the elimination of gravity. Furthermore, the experiments can be performed aboard the International Space Station (ISS) with essentially no solid container, since surface tension can provide liquid containment. Thus, the free surface can be readily observed (e.g., with high magnification objectives), manipulated (e.g., can be sheared or left otherwise undisturbed), and modified (e.g., by spreading a monomolecular oil film). First, we will perform experiments at 1g on a small droplet to ascertain the effect of forced convection on a liquid surface. Next, we will generalize our very successful computational model in annular systems to the spherical geometry of liquids at microgravity. We will design the space-based experiments based on the results from the ground experiments and numerical predictions. Finally, we will perform experiments aboard ISS on large-scale drops in an experiment referred to as the Ring-Sheared Drop (RSD).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	When proteins are perturbed from their native state, they lose their native confirmation (folding) and then can bond with other proteins, forming an aggregate. This is the reason why the transparent fluid in an egg becomes a white solid when cooked. Amyloid fibrils are a unique type of protein aggregate that occur naturally in the body. We study them because amyloid plaques in the brain are associated with many neurodegenerative disorders, including Alzheimer's. Fibrils are very intriguing because they are highly ordered and self-assemble. When heated, insulin molecules, for example, denature and bond with other denatured molecules (or assists in denaturing the other molecules). These form a nucleation site, and from there the fibril elongates as more molecule bond into what is called a "beta sheet" structure. The primary mechanism for activation of protein in the body is not yet understood, but it is likely to be a combination of flow friction, aka "shear stress" and interfacial effects that native proteins are exposed to. Aside from their central role in neurodegenerative diseases such as Parkinson's and Alzheimer's, formation of amyloid structures in proteins plays a role in other diseases including diabetes.
Task Progress:	Below is a list of major accomplishments during Year-3 of this project. A) A computational model was developed using the commercial package "COMSOL" for studying the formation and pinning of the liquid drops between two sharp-edged rings. B) An experiment was designed and conducted in collaboration with NASA Marshall Space Flight Center personnel, to test the formation and pinning of water drops using parabolic flights. Experiments were successfully conducted aboard ZeroG Corp. aircraft over a two day flight campaign in late February 2016. The results of the flight experiment along with comparisons to computational model predictions (item-A), as well as laboratory measurements along with corresponding simulations were used to prepare a manuscript. The manuscript was submitted for publication in the journal Microgravity Science and Technology. C) The driving of the flow by rotation of one ring in the RSD was simulated in the laboratory in an analog geometry known as the knife-edge surface viscometer. The main difference between the RSD and the knife-edge surface viscometer is that in the fluid is contained in a solid cylinder in the case of the viscometer as opposed to by surface tension as in the RSD. The experimental measurements in the knife-edge surface viscometer were reproduced by a numerical model developed from first principles at Arizona State University (ASU). The results were published in the Journal of Fluid Mechanics (Raghunandan et al., 2015). D) The role of fluid flow on the formation of amyloid fibrils at 65 degrees C was investigated in the laboratory using an annular flow device referred to as "rotating Couette" apparatus. The details of the flow field were analyzed using a computational model developed from first principles at ASU. The results were published in the journal Soft Matter (McBride et al., 2016). E) The fibrillization of protein at room temperature, i.e., similar to the RSD aboard the ISS, was studied in the laboratory in an annular flow device referred to as the deep-channel surface viscometer. This flow geometry is advantageous over the knife-edge surface viscometer with respect to the ease of imaging its surface with a microscope. The experiments revealed a never seen before mat of amyloid fibrils at the free surface. The measurements of the flow at the surface were interpreted using a computational model developed at ASU. The results have been submitted for publication in the journal Soft Matter (Balaraj et al., 2016). Article under review: Gulati S, Raghunandan A, Raheed F, McBride SA, and Hirsa AH. "Ring-Sheared Drop (RSD): Microgravity Module for Containerless Flow Studies," Microgravity Science & Technology (submitted). Article under review: Balaraj VS, Zeng P, Sanford SP, McBride SA, Raghunandan A., and Hirsa AH. "Surface Shear Viscosity as a Macroscopic Probe of Amyloid Fibril Formation at a Fluid Interface," Soft Matter (submitted).
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Articles in Peer-reviewed Journals	Raghunandan A, Lopez JM, Hirsa AH. "Bulk flow driven by a viscous monolayer." Journal of Fluid Mechanics. 2015 Dec;785:283-300. http://dx.doi.org/10.1017/jfm.2015.644 , Dec-2015
Articles in Peer-reviewed Journals	McBride SA, Sanford SP, Lopez JM, Hirsa AH. "Shear-induced amyloid fibrillization: the role of inertia." Soft Matter. 2016 Apr 14;12(14):3461-7. http://dx.doi.org/10.1039/c5sm02916c ; PubMed PMID: 26956731 , Apr-2016
Dissertations and Theses	Balaraj VS. "Study of Interfacial Rheology and Fibrillization Kinetics of Insulin in the Deep-Channel Surface Viscometer." MS Thesis, Rensselaer Polytechnic Institute, August 2016. , Aug-2016